

PHP Classes, Objects, and Forms

Classes and Objects

A **class** is a template for objects, and an **object** is an instance of a class.

Basic class definitions begin with the **class** keyword, followed by a class name, followed by a pair of curly braces, which enclose the definitions of the properties and methods belonging to the class.

The class name can be any valid label, provided it is not a PHP reserved word. A valid class name starts with a letter or underscore, followed by any number of letters, numbers, or underscores.

A class may contain its own constants, variables called **properties**, and functions called **methods**. Below is an example of a simple class definition with two (2) properties and two (2) methods.

```
<?php
class SimpleClass
{
    // property declaration
    public $name;

    // method declaration
    function set_name($name) {
        $this->name = $name;
    }
    function get_name() {
        return $this->name;
    }
}
?>
```

The **\$this** keyword refers to the current object and is only available inside methods.

To create an instance of a class, use the **new** keyword. Each instance/object created has the same properties and methods but can have different values. The following code shows how objects of the class named **SimpleClass** are created.

```
<?php
class SimpleClass
{
    public $name;

    function set_name($name) {
        $this->name = $name;
    }
    function get_name() {
        return $this->name;
    }
}
```

```
$name1 = new SimpleClass();
$name2 = new SimpleClass();
$name1->set_name('Fanny');
$name2->set_name('Lesley');

echo $name1->get_name();
echo "<br>";
echo $name2->get_name();
?>
```

Forms

A **form** represents a document section containing interactive controls for collecting user input.

To create a form, you use the **<form>** element with the **action** and **method** attributes.

```
<form action="register.php" method="post">
</form>
```

- **action** – specifies the URL that processes the form submission
- **method** – specifies the HTTP method for submitting the form. The most commonly used form methods are POST (invisible to others) and GET (visible to everyone).

The following example displays a simple HTML form with three (3) input fields and a submit button:

```
<html>
<body>

<form action="register.php" method="post">
Student No.: <input type="text" name="id"><br>
Name: <input type="text" name="name"><br>
Section: <input type="text" name="section"><br>
<input type="submit">
</form>

</body>
</html>
```

Once the user clicks the submit button, the form data is sent for processing to a PHP file named **register.php**. The form data is sent with the HTTP POST method.

To display the submitted data, simply echo the variables:

```
Welcome, <?php echo $_POST["name"]; ?><br>
```

Assuming the input name is *John*, the output will be:

```
Welcome, John
```

In the previous HTML form example, the input fields were optional. To make sure that a field will be filled out, declare an error variable to hold the error message for the required field. Then, add an if-else statement for the `$_POST` variable to check if the `$_POST` variable is empty. If it is empty, an error message is stored in the error variable; otherwise, it sends the user input data through the **test_input()** function.

```
<?php
// define the variables and set to empty values
$nameErr = "";
$name = "";

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    if (empty($_POST["name"])) {
        $nameErr = "Name is required";
    } else {
        $name = test_input($_POST["name"]);
    }
}
?>
```

To display the error message, add a script after each required field in the HTML form.

```
<form method="post" action="<?php echo htmlspecialchars($_SERVER["PHP_SELF"]);?>">

Name: <input type="text" name="name">
<span class="error">* <?php echo $nameErr;?></span>
<br><br>

</form>
```

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